

Telling the time

Early devices for telling the time depended on the regular burning of a candle, or the flow of water through a small hole. The first mechanical clocks used the regular rocking of a metal rod, called a foliot, to control the movement of a hand around a dial. Later clocks used pendulums, which swing back and forth. The movement is transferred to gears that drive the hands.



Small-scale model of Su Song's clock

Su Song's water clock

- **What?** Water clock
- **Who?** Su Song
- **Where and when?** China, around 1090

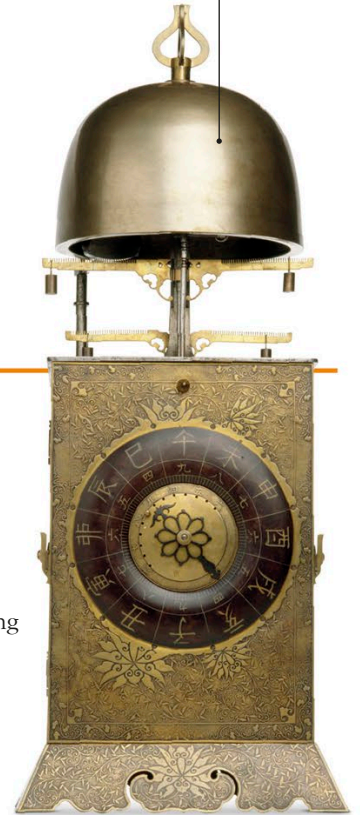
Water clocks were among the earliest devices for telling the time that did not depend on observation of the sun. Su Song's water clock, made in China in 1088, was one of the most complicated designs and was housed in a tower 35 ft (10 m) high. It was driven by a waterwheel and included 117 figures that came out of the tower on the hour and banged gongs.

Day and night clock

- **What?** Japanese lantern clock
- **Who?** Unknown
- **Where and when?** Japan, 19th century

Before about 1870, Japan divided both the day and the night into six equal hours. A daylight hour was a different length to a night hour and both varied according to the season. This clock has two timekeeping mechanisms, one for day and one for night.

It is called a Japanese lantern clock because it looks like a lantern.



19th-century Japanese lantern clock

The first watch

- **What?** Pocket watch
- **Who?** Peter Henlein
- **Where and when?** Germany, early 1500s

In the early 1500s, Peter Henlein, a locksmith from Nuremberg, Germany, miniaturized the large components of a clock into more portable devices, known as "pocket watches." These small, drumlike clocks were usually worn around the neck or attached to clothing—at this time they could not actually fit into a pocket. It was not until about a century later that smaller versions—the first true pocket watches—were designed.



A single hand marks the hours.

Early pocket watch